

Variational Calculus

Reading material: Chapter 6 of *Classical Mechanics* by John R. Taylor

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1. Single-Variable Functions

In ordinary differential calculus we study functions $f : \mathbb{R} \rightarrow \mathbb{R}$. To locate a local extremum or, more generally, a **stationary point** of f , we impose the first-order condition

$$\frac{df(x)}{dx} = 0. \tag{1}$$

At such a point the graph is **locally flat**: an infinitesimal change Δx produces no first-order change in f . Whether the stationary point is a minimum or a maximum must be decided by further analysis (e.g., second derivative test).

Calculus of variations asks the analogous question for an object that is *one level more complicated*: instead of a function of a number, we have a **functional**, i.e. a quantity that depends on an entire function. **We now seek the function $y(x)$ that makes the functional stationary.**

2. Functionals

A functional is a rule that takes an entire function as its input and returns a single number. For example, suppose $y(x)$ describes a path. A functional might return the length of that path, or the time it takes light to travel along that path.

In the problems we will study, the input function $y(x)$ is usually assumed to be sufficiently smooth. Our goal is to find the function $y(x)$ that makes the value of the functional stationary.

“Stationary” means that if we make a very small change to the function, the value of the functional does not change to first order. This is analogous to ordinary calculus, where a function $g(x)$ has a stationary point when $\frac{dg}{dx} = 0$. In calculus of variations, however, the variable is not a number x . The “variable” is an entire function $y(x)$.

A common type of functional has the form

$$F[y] = \int_{x_1}^{x_2} f(x, y(x), y'(x)) dx, \quad y' = \frac{dy}{dx}. \quad (2)$$

Here:

- $F[y]$ is the functional.
- $y(x)$ is the input function.
- $y'(x)$ is the derivative of the input function.
- $f(x, y, y')$ is an ordinary function of three variables.
- Once we choose a particular function $y(x)$, the integrand becomes an ordinary function of x , and the integral gives a number.

The square brackets in $F[y]$ remind us that the input is a function, not just a number.

In Lagrangian mechanics, the most important functional is the **action**,

$$S[q] = \int_{t_1}^{t_2} L(t, q(t), \dot{q}(t)) dt, \quad \dot{q} = \frac{dq}{dt}. \quad (3)$$

Here:

- $q(t)$ is the path of the system in time.
- $\dot{q}(t)$ is the velocity.
- $L(t, q, \dot{q})$ is the Lagrangian.
- $S[q]$ is a number assigned to the entire path $q(t)$.

Hamilton's principle says that the physically realized path $q(t)$ makes the action stationary:

$$\delta S = 0 \quad (4)$$

This does **not** always mean that the action is minimized. It means that the first-order change in the action vanishes.

3. Examples of Functionals

A functional is a rule that takes in an entire function and returns a single number.

For example, is

$$F(y) = (y(x))^2$$

it a functional?

No. If $y(x) = x^2$, then

$$F(y) = x^4,$$

which is still a function of x , not a single number.

However,

$$F[y] = \int_0^1 (y(x))^2 dx$$

is a functional. If $y(x) = x^2$, then

$$F[y] = \int_0^1 x^4 dx = \frac{1}{5}.$$

This time, the output is a single number.

3.1 Length of a Plane Curve

Suppose a curve in a two-dimensional plane is described by $y = y(x)$ between $x = x_1$ and $x = x_2$. The arc length element is

$$ds = \sqrt{dx^2 + dy^2}.$$

Since

$$dy = y'(x) dx,$$

we get

$$ds = \sqrt{1 + y'(x)^2} dx.$$

Therefore the total length of the curve is

$$L[y] = \int_{x_1}^{x_2} \sqrt{1 + y'(x)^2} dx. \quad (5)$$

This is a functional because the input is the entire curve $y(x)$, and the output is the total length, a number.

If the endpoints are fixed,

$$y(x_1) = y_1, \quad y(x_2) = y_2,$$

then the problem is:

Among all smooth curves connecting the two endpoints, which one has the shortest length?

The answer is a straight line. Calculus of variations provides a systematic way to prove this.

3.2 Fermat's Principle

Light traveling through a medium with refractive index $n(x, y)$ has speed

$$v = \frac{c}{n(x, y)}.$$

For a small distance ds , the travel time is

$$dt = \frac{ds}{v} = \frac{n(x, y)}{c} ds.$$

Using

$$ds = \sqrt{1 + y'(x)^2} dx,$$

the total travel time is

$$T[y] = \frac{1}{c} \int_{x_1}^{x_2} n(x, y) \sqrt{1 + y'(x)^2} dx. \quad (6)$$

Fermat's principle states that light traveling between two points follows a path for which the travel time is extremal, typically minimal. Equivalently, the actual path taken by light is a stationary path of the travel-time functional:

$$\delta T = 0.$$

If n is constant, then

$$T[y] = \frac{n}{c} \int_{x_1}^{x_2} \sqrt{1 + y'(x)^2} dx.$$

So minimizing travel time is equivalent to minimizing path length. Therefore, in a uniform medium, light travels in a straight line.

If n varies with position, then the fastest path is generally not a straight line. Calculus of variations is the tool that determines the actual ray path.

4. Variation

We now want to define the analogue of a derivative for a functional.

In ordinary calculus, to study a function $g(x)$, we look at what happens when we change x slightly:

$$x \rightarrow x + \varepsilon.$$

For a functional $F[y]$, the input is not a number. The input is a whole function $y(x)$. So instead of changing a number slightly, we change the whole function slightly.

We introduce a nearby function

$$y_\varepsilon(x) = y(x) + \varepsilon \eta(x).$$

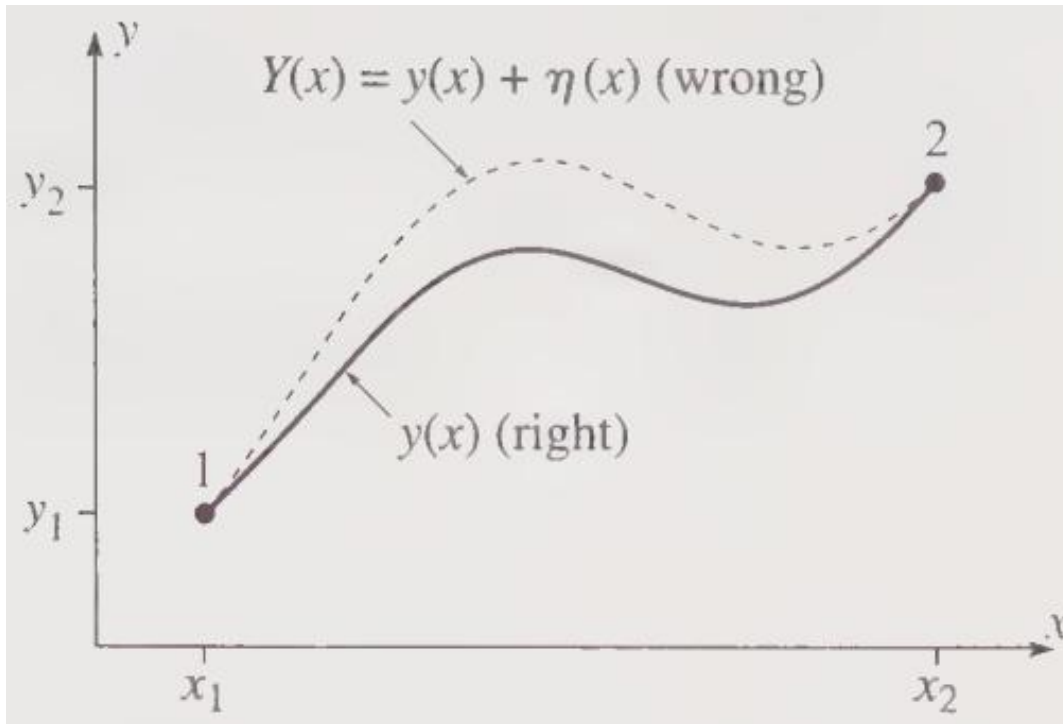


Figure 1: The path $y = y(x)$ between points 1 and 2 is the "right" path. Any other path $Y(x) = y(x) + \eta(x)$ is a variation that gives a larger value for the functional.

Here:

- $y(x)$ is the original function.
- $\eta(x)$ is a chosen "shape" of the perturbation.
- ε is a small real number that controls the size of the perturbation.
- $y_\varepsilon(x)$ is the perturbed function.

The function $\eta(x)$ tells us **which direction in function space** we are moving away from $y(x)$.

The number ε tells us **how far** we move in that direction.

So $\eta(x)$ plays a role similar to a direction vector, while ε plays a role similar to a small step size.

4.1 What Does "Direction in Function Space" Mean?

In ordinary two-dimensional space, if we start at a point $\mathbf{r}$, we can move in the direction of a vector $\mathbf{v}$:

$$\mathbf{r}_\varepsilon = \mathbf{r} + \varepsilon \mathbf{v}.$$

Here:

- $\mathbf{v}$ gives the direction.
- ε gives the step size.

Similarly, in function space, we write

$$y_\varepsilon(x) = y(x) + \varepsilon \eta(x).$$

Here:

- $\eta(x)$ gives the "direction" in function space.
- ε gives the size of the step.

For example, if

$$y(x) = x^2$$

and

$$\eta(x) = \sin(\pi x),$$

then

$$y_\varepsilon(x) = x^2 + \varepsilon \sin(\pi x).$$

For small ε , this is a curve very close to $y(x) = x^2$, but slightly deformed in the shape of $\sin(\pi x)$.

If $\varepsilon > 0$, we move in the $+\eta$ direction.

If $\varepsilon < 0$, we move in the $-\eta$ direction.

4.2 Fixed Endpoints

Often we want **all allowed curves to pass through the same two endpoints**:

$$\eta(x_1) = 0, \quad \eta(x_2) = 0.$$

Thus, the allowed variations must vanish at the endpoints.

This means that the perturbation can change the curve in the middle, but not at the two ends.

4.3 The First Variation

In ordinary calculus, suppose g is a function of one variable, x . If we change x by a small amount ε , then $g(x)$ changes to $g(x + \varepsilon)$. The rate of change of g with respect to x is

$$\frac{dg}{dx} = \left. \frac{d}{d\varepsilon} g(x + \varepsilon) \right|_{\varepsilon=0}.$$

In words, this says:

Start at x , move a small amount ε , compute the change in g , and then look at the instantaneous rate of change at $\varepsilon = 0$.

The **first variation** of F at y in the direction η is defined by

$$\delta F[y; \eta] = \left. \frac{d}{d\varepsilon} F[y + \varepsilon \eta] \right|_{\varepsilon=0}. \quad (7)$$

In words, this means

Start with the function y , perturb it slightly in the direction η by an infinitesimal amount ε , compute the resulting change in the functional F , and keep only the **first-order**, or **linear**, part of that change.

$$F[y + \varepsilon \eta] = F[y] + \varepsilon \delta F[y; \eta] + \text{higher-order terms}$$

This is directly analogous to the first derivative in ordinary calculus.

Here, x is replaced by the function y , and the small change ε is replaced by a small change in the function, $\varepsilon\eta$.

So the first variation measures how the functional changes when we move slightly away from the function $y(x)$ in the direction $\eta(x)$.

Note. There is a subtle notational difference between ordinary calculus and variational calculus. In ordinary calculus, $\frac{dg}{dx}$, is the rate of change of g in the x -direction. If x is perturbed by a small amount ε , then the first-order change in g is $\varepsilon \frac{dg}{dx}$. In variational calculus, though $\delta F[y; \eta]$ is called the first variation, it is technically the directional derivative, or rate of change of the functional F at y in the direction η . If y is perturbed to $y + \varepsilon\eta$, then the first-order change in F is $\varepsilon \delta F[y; \eta]$. Thus, strictly speaking, $\delta F[y; \eta]$ is the directional derivative of F , while $\varepsilon \delta F[y; \eta]$ is the corresponding first-order change in F .

4.4 Variation and Variation of the Derivative

Because

$$y_\varepsilon(x) = y(x) + \varepsilon\eta(x),$$

Then the change in y is

$$\Delta y(x) = y_\varepsilon(x) - y(x) = \varepsilon\eta(x).$$

The **variation** is defined as the derivative with respect to ε at $\varepsilon = 0$:

$$\delta y(x) = \left. \frac{d}{d\varepsilon} y_\varepsilon(x) \right|_{\varepsilon=0} = \lim_{\varepsilon \rightarrow 0} \frac{y_\varepsilon(x) - y_0(x)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{\Delta y(x)}{\varepsilon} = \eta(x). \quad (8)$$

We also have

$$y'_\varepsilon(x) = y'(x) + \varepsilon\eta'(x).$$

Similarly,

$$\delta y'(x) = \left. \frac{d}{d\varepsilon} y'_\varepsilon(x) \right|_{\varepsilon=0} = \eta'(x). \quad (9)$$

Therefore,

$$\delta y'(x) = \frac{d}{dx} \delta y(x). \quad (10)$$

This is allowed because **the variation changes the function y , but not the independent variable x .**

In other words, x is held fixed while the curve is varied.

4.5 Stationary Function

A function $y(x)$ is called a **stationary function** of the functional $F[y]$ if

$$\delta F[y; \eta] = 0 \quad (11)$$

for every allowed perturbation $\eta(x)$.

For fixed endpoints, this means

$$\delta F[y; \eta] = 0 \quad \text{for all smooth } \eta(x) \text{ with } \eta(x_1) = \eta(x_2) = 0. \quad (12)$$

This is the variational analogue of the ordinary calculus condition $\frac{dg}{dx} = 0$.

Similarly, in calculus of variations, a stationary function may correspond to a minimum, a maximum, or a saddle point in function space. Determining which one it is usually requires further analysis, such as studying the second variation.

4.6 Summary

A functional takes a function as input and returns a number:

$$y(x) \mapsto F[y].$$

To study how a functional changes, we perturb the input function:

$$y(x) \rightarrow y_\varepsilon(x) = y(x) + \varepsilon \eta(x).$$

Here:

- $\eta(x)$ is the shape or direction of the perturbation.
- ε is a small parameter controlling the size and sign of the perturbation.
- Fixed endpoints require

$$\eta(x_1) = \eta(x_2) = 0.$$

The first variation is

$$\delta F[y; \eta] = \left. \frac{d}{d\varepsilon} F[y + \varepsilon \eta] \right|_{\varepsilon=0}.$$

A stationary function satisfies

$$\delta F[y; \eta] = 0$$

for every allowed variation $\eta(x)$.

This is the key idea behind the calculus of variations and leads directly to the Euler-Lagrange equation.

5. Euler-Lagrange Equation

For a functional of the form

$$F[y] = \int_{x_1}^{x_2} f(x, y(x), y'(x)) \, dx, \quad (13)$$

with fixed endpoints $y(x_1) = y_1$ and $y(x_2) = y_2$, the functional F is said to be **stationary at** a twice-differentiable function $y(x)$ if its first variation vanishes for all admissible variations preserving the endpoints:

$$\delta F[y] = 0. \quad (14)$$

If F is stationary at $y(x)$, then $y(x)$ must satisfy the **Euler-Lagrange equation**:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0. \quad (15)$$

Here, $\partial f / \partial y$ and $\partial f / \partial y'$ denote the partial derivatives of the integrand $f(x, y, y')$ with respect to its second and third arguments, respectively, evaluated along the path $y = y(x)$.

If the functional depends on several dependent variables,

$$F[y, z] = \int_{x_1}^{x_2} f(x, y, y', z, z') dx, \quad (16)$$

then each dependent variable yields its own Euler-Lagrange equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0, \quad \frac{\partial f}{\partial z} - \frac{d}{dx} \frac{\partial f}{\partial z'} = 0. \quad (17)$$

5.1 Derivation of the Euler-Lagrange Equation

Consider the functional

$$F[y] = \int_{x_1}^{x_2} f(x, y(x), y'(x)) dx, \quad (18)$$

with fixed boundary values $y(x_1)$ and $y(x_2)$. We want to find the condition that a stationary function $y(x)$ must satisfy.

To test whether $y(x)$ is stationary, we compare it with nearby functions. Introduce the perturbed function

$$y_\varepsilon(x) = y(x) + \varepsilon \eta(x),$$

where $\eta(x)$ is a smooth function satisfying $\eta(x_1) = \eta(x_2) = 0$, which ensure that the endpoints remain fixed. Previously, we defined the first variation by

$$\delta F[y; \eta] = \left. \frac{d}{d\varepsilon} F[y + \varepsilon \eta] \right|_{\varepsilon=0}. \quad (19)$$

Here $F[y + \varepsilon \eta]$ means the value of the functional evaluated on the perturbed function $y_\varepsilon(x)$. We compute $F[y + \varepsilon \eta]$ by replacing $y(x)$ with $y_\varepsilon(x)$, and replacing $y'(x)$ with $y'_\varepsilon(x)$.

Now,

$$y_\varepsilon(x) = y(x) + \varepsilon \eta(x),$$

so

$$y'_\varepsilon(x) = \frac{d}{dx}[y(x) + \varepsilon\eta(x)] = y'(x) + \varepsilon\eta'(x).$$

Therefore,

$$F[y + \varepsilon\eta] = F[y_\varepsilon] = \int_{x_1}^{x_2} f(x, y + \varepsilon\eta, y' + \varepsilon\eta') dx. \quad (20)$$

This formula is simply the original definition of the functional, evaluated on the nearby function $y_\varepsilon(x)$.

At this point, $F[y + \varepsilon\eta]$ is an ordinary function of the single real variable ε . We can therefore differentiate it with respect to ε . Assuming f is sufficiently smooth, we may differentiate under the integral sign:

$$\delta F[y; \eta] = \left. \frac{d}{d\varepsilon} \int_{x_1}^{x_2} f(x, y + \varepsilon\eta, y' + \varepsilon\eta') dx \right|_{\varepsilon=0}.$$

Thus,

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \left. \frac{d}{d\varepsilon} f(x, y + \varepsilon\eta, y' + \varepsilon\eta') \right|_{\varepsilon=0} dx.$$

Now use the ordinary chain rule. The function f depends on three arguments:

$$f = f(x, y, y').$$

When ε changes, x does not change. However, the second and third arguments do change:

$$y \rightarrow y + \varepsilon\eta,$$

and

$$y' \rightarrow y' + \varepsilon\eta'.$$

Therefore,

$$\frac{d}{d\varepsilon} f(x, y + \varepsilon\eta, y' + \varepsilon\eta') = \frac{\partial f}{\partial y}\eta + \frac{\partial f}{\partial y'}\eta'.$$

Setting $\varepsilon = 0$, we get the first variation

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\eta + \frac{\partial f}{\partial y'}\eta' \right) dx \Big|_{\varepsilon=0}. \quad (21)$$

Here the partial derivatives are evaluated along the original function $y(x)$, meaning

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial y}(x, y(x), y'(x)),$$

and

$$\frac{\partial f}{\partial y'} = \frac{\partial f}{\partial y'}(x, y(x), y'(x)).$$

So we have

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\eta + \frac{\partial f}{\partial y'}\eta' \right) dx. \quad (22)$$

Another common way to write this is

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial y'}\delta y' \right) dx. \quad (23)$$

This notation is analogous to ordinary differential calculus. For a function $f(x, y)$, the total differential is

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy. \quad (24)$$

Similarly, in the calculus of variations, the first variation δF describes the first-order change in the functional $F[y]$ caused by a small change δy in the function y .

The first term already contains η , but the second term contains η' . We want to rewrite the expression so that the variation η appears without its derivative. To do this, we integrate the second term by parts:

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y'} \eta' dx = \left. \frac{\partial f}{\partial y'} \eta \right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) dx. \quad (25)$$

Recall

$$\int_a^b uv' dx = \int_a^b u dv = (uv)|_a^b - \int_a^b v du \quad (26)$$

The boundary term vanishes because $\eta(x_1) = \eta(x_2) = 0$. Therefore, $\left. \frac{\partial f}{\partial y'} \eta \right|_{x_1}^{x_2} = 0$.

Hence the first variation becomes

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \eta(x) \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] dx. \quad (27)$$

For y to be stationary, this integral must be zero for *every* admissible η . The **fundamental lemma (引理) of the calculus of variations** then implies that the factor multiplying η must vanish everywhere on $[x_1, x_2]$:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0. \quad (28)$$

5.2 Proof of the Fundamental Lemma

Let $g(x)$ be continuous on $[x_1, x_2]$, and suppose

$$\int_{x_1}^{x_2} \eta(x) g(x) dx = 0$$

for every smooth function $\eta(x)$ satisfying

$$\eta(x_1) = \eta(x_2) = 0.$$

We want to show that this forces

$$g(x) = 0$$

for all $x \in [x_1, x_2]$.

Suppose, for contradiction, that g is not identically zero. Then there is some point x_0 where

$$g(x_0) \neq 0.$$

Since g is continuous, it must remain nonzero with the same sign on some small interval around x_0 .

For example, suppose $g(x) > 0$ on a small interval. Then we can choose a smooth function $\eta(x)$ that is positive on that interval and zero outside it, while still satisfying

$$\eta(x_1) = \eta(x_2) = 0.$$

For this choice, the product $\eta(x)g(x)$ is nonnegative everywhere and positive on part of the interval. Therefore,

$$\int_{x_1}^{x_2} \eta(x)g(x) dx > 0,$$

which contradicts the assumption that the integral is zero for every η .

Similarly, if $g(x) < 0$ on a small interval, we choose $\eta(x) < 0$ there, so that $\eta(x)g(x) > 0$ on that interval. Again the integral would be positive, giving a contradiction.

Therefore, g must be identically zero:

$$g(x) = 0 \quad \text{for all } x \in [x_1, x_2].$$

Applying this result to

$$g(x) = \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right),$$

we obtain

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0.$$

Thus the Euler-Lagrange equation is a necessary condition for a stationary function.

6. The Beltrami Identity

When the integrand does not depend explicitly on x , i.e.

$$f = f(y, y'), \tag{29}$$

the Euler-Lagrange equation can be integrated once. This first integral is called the **Beltrami identity** (or, in physics, it is related to energy conservation when the independent variable is time).

Assume both f and y are C^2 (twice continuously differentiable). Along the path,

$$\frac{df}{dx} = \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y''. \tag{30}$$

Because $\partial f / \partial x = 0$. Using the Euler-Lagrange equation to replace $\partial f / \partial y$ by $\frac{d}{dx} (\partial f / \partial y')$, we get

$$\frac{df}{dx} = \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) y' + \frac{\partial f}{\partial y'} y'' = \frac{d}{dx} \left(y' \frac{\partial f}{\partial y'} \right). \tag{31}$$

Hence

$$\frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) = 0, \tag{32}$$

and therefore

$$f - y' \frac{\partial f}{\partial y'} = C, \quad (33)$$

where C is a constant of integration. This is a first-order differential equation, often much easier to solve than the second-order Euler–Lagrange equation.

7. Application of Variational Calculus

The general procedure for solving a variational problem is:

1. **Write the functional.** Express the quantity of interest (length, time, action, etc.) as an integral

$$F[y] = \int_{x_1}^{x_2} f(x, y, y') dx. \quad (34)$$

2. **Identify the integrand.** Read off the function $f(x, y, y')$.
3. **Apply the Euler–Lagrange equation.** Compute the partial derivatives and solve

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0. \quad (35)$$

If f has no explicit x -dependence, you may use the Beltrami identity instead.

4. **Solve the differential equation** subject to the boundary conditions $y(x_1) = y_1, y(x_2) = y_2$. The resulting function is the stationary path.
5. **Check the nature of the stationary path** if a minimum or maximum is required. The Euler–Lagrange equation by itself guarantees only stationarity.

8. Shortest Path between Two Points

We can now settle the very first example of these notes (Section 3.1): among all smooth curves $y(x)$ joining two fixed points 1 and 2, the one with the shortest length is a straight line.

From Section 3.1, the length of the path is

$$L[y] = \int_{x_1}^{x_2} \sqrt{1 + y'^2} dx, \quad (36)$$

which is a functional of the standard form with the integrand

$$f(y, y') = (1 + y'^2)^{1/2}. \quad (37)$$

Because this integrand has no explicit x -dependence, we can solve the problem with the **Beltrami identity**. We need only the derivative

$$\frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}. \quad (38)$$

The Beltrami identity gives

$$f - y' \frac{\partial f}{\partial y'} = \sqrt{1 + y'^2} - \frac{y'^2}{\sqrt{1 + y'^2}} = \frac{1}{\sqrt{1 + y'^2}} = C, \quad (39)$$

where C is a constant of integration. Inverting this result,

$$1 + y'^2 = \frac{1}{C^2}, \quad (40)$$

so y'^2 is constant. Hence $y'(x) = m$ for some constant m , and integrating once more gives

$$\boxed{y(x) = mx + b} \quad (41)$$

— the shortest path between two points is a straight line.

(The same conclusion can be reached from the Euler–Lagrange equation; see Sections 5–5.1.)

9. Brachistochrone Problem

The **brachistochrone problem** (最速降线问题) asks: given two points 1 and 2 in a vertical plane, with 1 higher above the ground, in what shape should we build a frictionless track so that a bead (or roller-coaster car) released from rest at point 1 reaches point 2 in the shortest possible time? The name comes from the Greek words *brachistos* (shortest) and *chronos* (time).

We take point 1 as the origin and choose to measure y vertically **downward**. The time to travel from 1 to 2 is

$$\text{time}(1 \rightarrow 2) = \int_1^2 \frac{ds}{v}, \quad (42)$$

where the speed at any depth y is fixed by conservation of energy:

$$\frac{1}{2}mv^2 = mgy \implies v = \sqrt{2gy}. \quad (43)$$

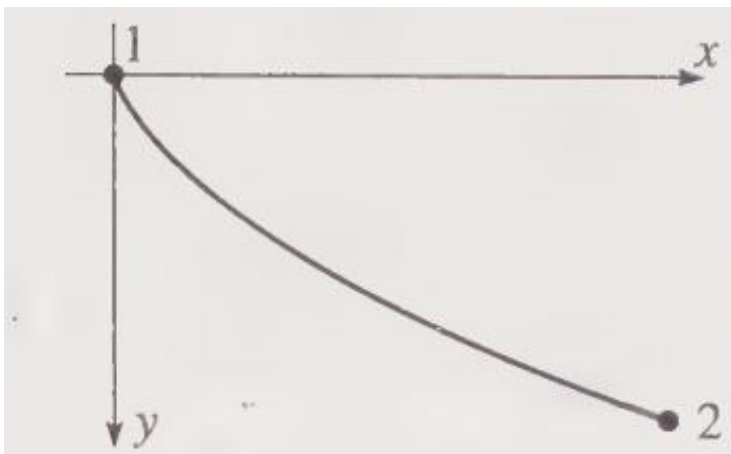


Figure 2: The brachistochrone problem is to find the shape of track on which a roller coaster released from point 1 will reach point 2 in the minimum possible time.

In this problem the independent variable is y , not x . Because the speed $v = \sqrt{2gy}$ is a function of y alone, it is convenient to write the unknown path as $x = x(y)$, with y playing the role that x played before. A prime now denotes differentiation with respect to y : $x'(y) = dx/dy$. The arc-length element must therefore be written as

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{x'(y)^2 + 1} dy, \quad (44)$$

and the travel time becomes

$$\text{time}(1 \rightarrow 2) = \frac{1}{\sqrt{2g}} \int_0^{y_2} \frac{\sqrt{x'(y)^2 + 1}}{\sqrt{y}} dy. \quad (45)$$

This is of the standard form, with the roles of x and y interchanged, and the integrand

$$f(x, x', y) = \frac{\sqrt{x'^2 + 1}}{\sqrt{y}}. \quad (46)$$

Applying the Euler–Lagrange equation with the variables swapped,

$$\frac{\partial f}{\partial x} - \frac{d}{dy} \frac{\partial f}{\partial x'} = 0. \quad (47)$$

Since f does not depend on x , this says simply that $\partial f / \partial x'$ is a constant. Evaluating the derivative (and squaring for convenience),

$$\frac{x'^2}{y(1 + x'^2)} = \text{const} = \frac{1}{2a}, \quad (48)$$

where the constant has been named $1/2a$ for future convenience. Solving for x' ,

$$x' = \sqrt{\frac{y}{2a - y}}, \quad \text{so} \quad x = \int \sqrt{\frac{y}{2a - y}} dy. \quad (49)$$

This integral is evaluated by the unlikely-looking substitution

$$y = a(1 - \cos \theta), \quad (50)$$

which gives (as you should check)

$$x = a \int (1 - \cos \theta) d\theta = a(\theta - \sin \theta) + \text{const}. \quad (51)$$

The starting point 1 is at $x = y = 0$, which corresponds to $\theta = 0$, so the integration constant is zero. The stationary path is therefore the curve given parametrically by

$$\boxed{x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta)} \quad (52)$$

with the constant a chosen so that the curve passes through the prescribed endpoint (x_2, y_2) .

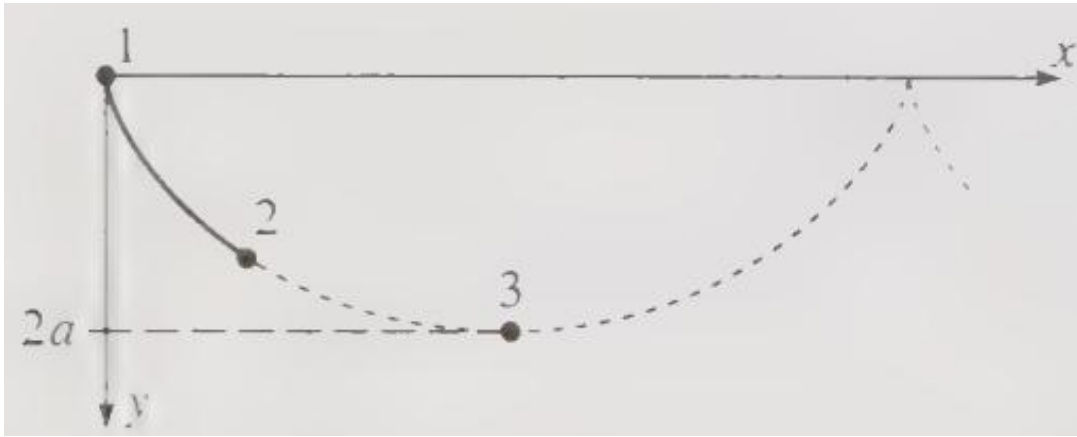


Figure 3: The path that gives the shortest time between the given points 1 and 2 is part of a cycloid, with a vertex at 1 and passing through 2. The cycloid is the curve traced by a point on the rim of a wheel of radius a rolling along the underside of the x -axis. Point 3 is the lowest point on the curve.

The resulting curve is a **cycloid (摆线)** — the curve traced out by a point on the rim of a wheel of radius a rolling along the underside of the x -axis.

A remarkable feature of this curve: if the cart is released from rest at point 2 and rolls to the bottom of the curve (point 3), the time to roll from 2 to 3 is the same whatever the position of 2. The oscillations of a cart rolling back and forth on a cycloid-shaped track are therefore *exactly isochronous* (period independent of amplitude) — unlike a simple pendulum, which is only approximately so. This property was actually used in the design of some historical clocks.

10. More than Two Variables

So far we have considered only problems with two variables: one independent (usually x) and one dependent (usually y). For most applications in mechanics there are **several dependent variables**, though still only one independent variable, usually the time t .

A simple example where a second dependent variable is forced on us: the shortest path between two points. We assumed the path could be written as $y = y(x)$. Reasonable as this seems, it is easy to think of paths that cannot be written this way, such as the path below, which doubles back on itself. To include all possible paths, we write the path in **parametric form**

$$x = x(u), \quad y = y(u), \quad (53)$$

where u is any convenient parameter (for instance, the distance along the path).

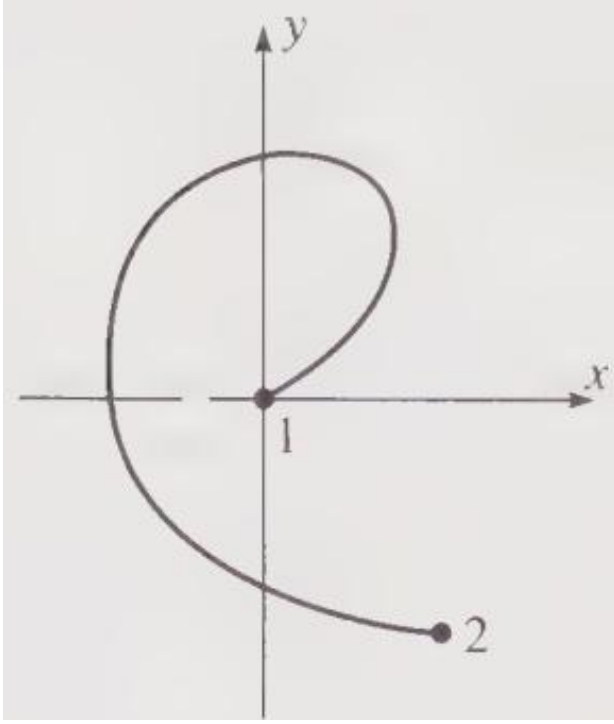


Figure 4: This path between the two points 1 and 2 cannot be written as $y = y(x)$ nor as $x = x(y)$, but it can be written in parametric form.

The length of the path is then

$$L = \int_{u_1}^{u_2} \sqrt{x'(u)^2 + y'(u)^2} du, \quad (54)$$

and our job is to find the two functions $x(u)$ and $y(u)$ for which this integral is stationary. The general problem of this type is

$$S = \int_{u_1}^{u_2} f(x(u), y(u), x'(u), y'(u), u) du \quad (55)$$

between two fixed endpoints. The solution parallels the one-variable case, but now we perturb *both* functions. Let the correct path be $x(u)$, $y(u)$, and consider a neighboring "wrong" path

$$x(u) + \alpha \xi(u), \quad y(u) + \beta \eta(u), \quad (56)$$

where ξ and η are arbitrary functions vanishing at the endpoints. The requirement that S be stationary at $\alpha = \beta = 0$ then yields, by exactly the argument of Section 5.1, **one Euler-Lagrange equation per dependent variable**:

$$\frac{\partial f}{\partial x} - \frac{d}{du} \frac{\partial f}{\partial x'} = 0, \quad \frac{\partial f}{\partial y} - \frac{d}{du} \frac{\partial f}{\partial y'} = 0. \quad (57)$$

Example (shortest path, again). Here $f = \sqrt{x'^2 + y'^2}$ is independent of x and y , so both equations say that $\partial f / \partial x'$ and $\partial f / \partial y'$ are constants. Dividing one by the other,

$$\frac{dy}{dx} = \frac{y'}{x'} = \text{constant} = m, \quad (58)$$

so the required path is a straight line $y = mx + b$ — now proved for *all* possible paths, including those that cannot be written as $y = y(x)$.